

CYBERSNOVA



PROJECT REPORT

Prepared by
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Field	Details
Full Name	Namrata Ravindra Pawar
Registration Number	CSN-SP-2026-155
Internship Domain	Cyber Security
Duration	1 month
Start Date	1 Sept 2026
End Date	30 Sept 2026
Mode	30 Sept 2026
Assigned Mentor	Internship Coordinator

Level 1=Intermediate

Level

**PROJECT 2: PORTSWIGGER WEB
SECURITY ACADEMY LABS**

Introduction

The PortSwigger Web Security Academy Lab Project was conducted to gain practical experience in identifying and exploiting common web application security vulnerabilities in a controlled and legally authorized environment. The Web Security Academy provides interactive labs containing deliberately vulnerable web applications that allow students to practice real-world web security testing techniques.

As part of this project, five Apprentice-level labs were successfully completed. The selected labs covered different vulnerability classes, including SQL Injection, Cross-Site Scripting (XSS), Broken Access Control, and File Path Traversal.

The practical exercises involved analyzing application behavior, identifying vulnerable input points, crafting appropriate test payloads, using Burp Suite for HTTP request analysis and modification, and verifying the impact of each vulnerability. The project also focused on understanding appropriate defensive measures that can be implemented to prevent similar vulnerabilities in real-world applications.

Objective

The main objectives of this project were:

- To understand common web application vulnerabilities through practical laboratory exercises.
- To identify vulnerable parameters, inputs, and application functionality.
- To learn basic techniques for testing SQL Injection vulnerabilities.
- To understand and demonstrate Reflected Cross-Site Scripting (XSS).
- To identify and exploit improperly protected administrative functionality.
- To understand the risks associated with File Path Traversal vulnerabilities.
- To use Burp Suite to intercept, inspect, and modify HTTP requests.
- To analyze the security impact of successful exploitation.
- To understand appropriate defensive measures for the identified vulnerabilities.
- To document the exploitation process using screenshots and technical explanations.
- To successfully complete five Apprentice-level PortSwigger Web Security Academy labs and provide proof of completion.

Tools and Environment

The following tools and environment were used during the assessment:

- **Web Security Training Platform**

PortSwigger Web Security Academy was used as the authorized training environment. The platform provides interactive labs designed to demonstrate and practice web application security vulnerabilities.

- **Web Browser**

A web browser was used to access the PortSwigger lab applications, interact with vulnerable functionality, submit test inputs, and verify successful exploitation.

- **Burp Suite Community Edition**

Burp Suite was used as the primary web security testing tool. It was used to inspect HTTP requests and responses and, where required, modify requests to demonstrate vulnerabilities.

- **Vulnerability Categories Tested**

The project covered the following vulnerability categories:

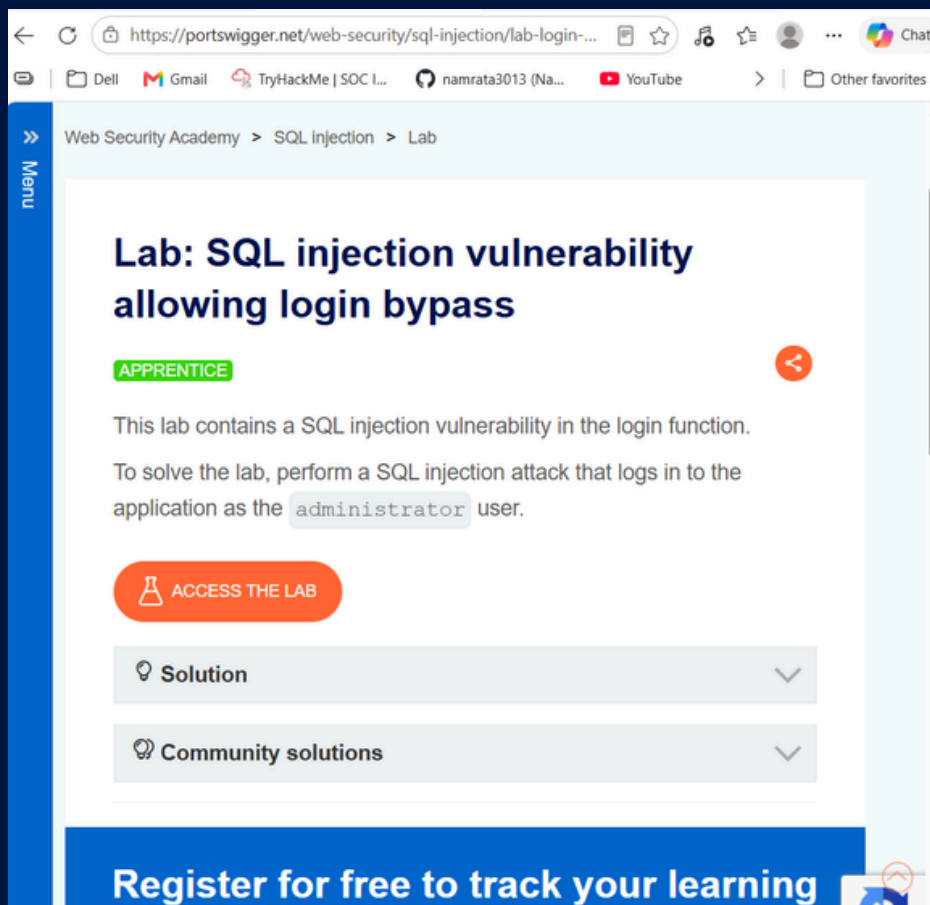
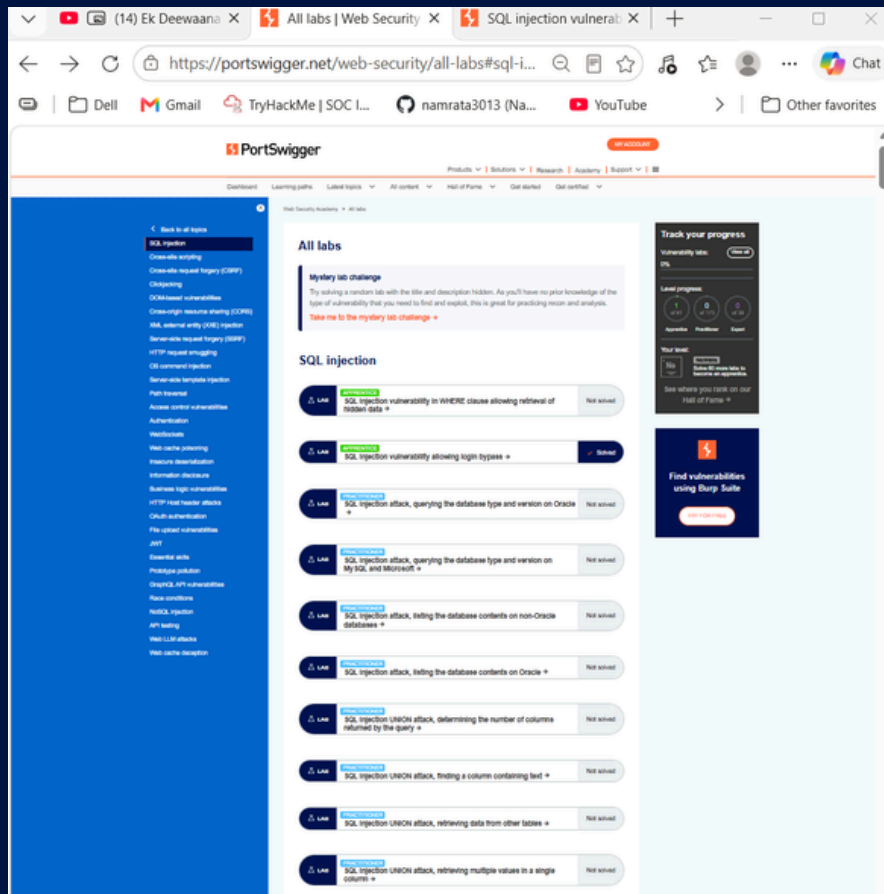
1. **SQL Injection**
2. **Reflected Cross-Site Scripting (XSS)**
3. **Broken Access Control / Unprotected Administrative Functionality**
4. **File Path Traversal**

PortSwigger's Web Security Academy includes dedicated learning material and labs for these vulnerability classes

- **Testing Environment**

The testing was performed against PortSwigger's intentionally vulnerable lab instances. Testing was limited to the authorized laboratory targets provided by the Web Security Academy.

Lab 1 – SQL Injection: Login Bypass



WebSecurity Academy SQL injection vulnerability allowing login bypass LAB Not solved

Back to lab description >>

Home | My account

WE LIKE TO SHOP



The Giant Enter Key
★★★★★ \$75.74



High-End Gift Wrapping
★★★★★ \$40.27



More Than Just Birdsong
★★★★★ \$37.43



Beat the Vacation Traffic
★★★★★ \$73.48

WebSecurity Academy SQL injection vulnerability allowing login bypass LAB Solved

Back to lab description >>

Congratulations, you solved the lab! Share your skills! Continue learning >>

Home | My account | Log out

My Account

Your username is: administrator
Your email is: namrata30pawar3@gmail.com

Email

Update email

Lab 2 – SQL Injection: Retrieval of Hidden Data

The screenshot shows a web browser at <https://portswigger.net/web-security/sql-injection/lab-r...>. The page is titled "Academy home" and "Web Security Academy > SQL Injection > Lab". The main heading is "Lab: SQL injection vulnerability in WHERE clause allowing retrieval of hidden data". It is labeled "APPRENTICE" and "LAB Not solved". The description states: "This lab contains a SQL injection vulnerability in the product category filter. When the user selects a category, the application carries out a SQL query like the following:" followed by a code block: `SELECT * FROM products WHERE category = 'Gifts'`. Below the code, it says: "To solve the lab, perform a SQL injection attack that causes the application to display one or more unreleased products." There is an "ACCESS THE LAB" button and sections for "Solution" and "Community solutions".

The screenshot shows the Web Security Academy lab interface. The top navigation bar includes the "Web Security Academy" logo and the lab title "SQL injection vulnerability in WHERE clause allowing retrieval of hidden data". A "LAB Not solved" status is shown. Below the title, there is a "Back to lab description" link. The main content area displays a "SHOP" application with the text "WE LIKE TO SHOP" and a shopping cart icon. A search bar is present with the text "Refine your search:" and a list of categories: "All", "Clothing, shoes and accessories", "Gifts", "Pets", and "Tech gifts". The bottom of the page features a row of four images: a silhouette of a person running, a hospital corridor, a person in a Santa hat, and a train.

Web Security Academy SQL injection vulnerability in WHERE clause allowing retrieval of hidden data LAB Not solved

Back to lab home Back to lab description >>

Home

WE LIKE TO SHOP

Gifts

Refine your search: All Clothing, shoes and accessories Gifts Pets Tech gifts

Snow Delivered To Your Door

High-End Gift Wrapping

Couple's Umbrella

Web Security Academy SQL injection vulnerability in WHERE clause allowing retrieval of hidden data LAB Solved

Back to lab description >>

Congratulations, you solved the lab! Share your skills! Continue learning >>

Home

WE LIKE TO SHOP

Gifts' OR 1=1--

Refine your search: All Clothing, shoes and accessories Gifts Pets Tech gifts

Fur Babies

AbZorba Ball

Picture Box

Conversation Controlling Lemon

Hologram Stand in

More Than Just Birdsong

Potato Theater

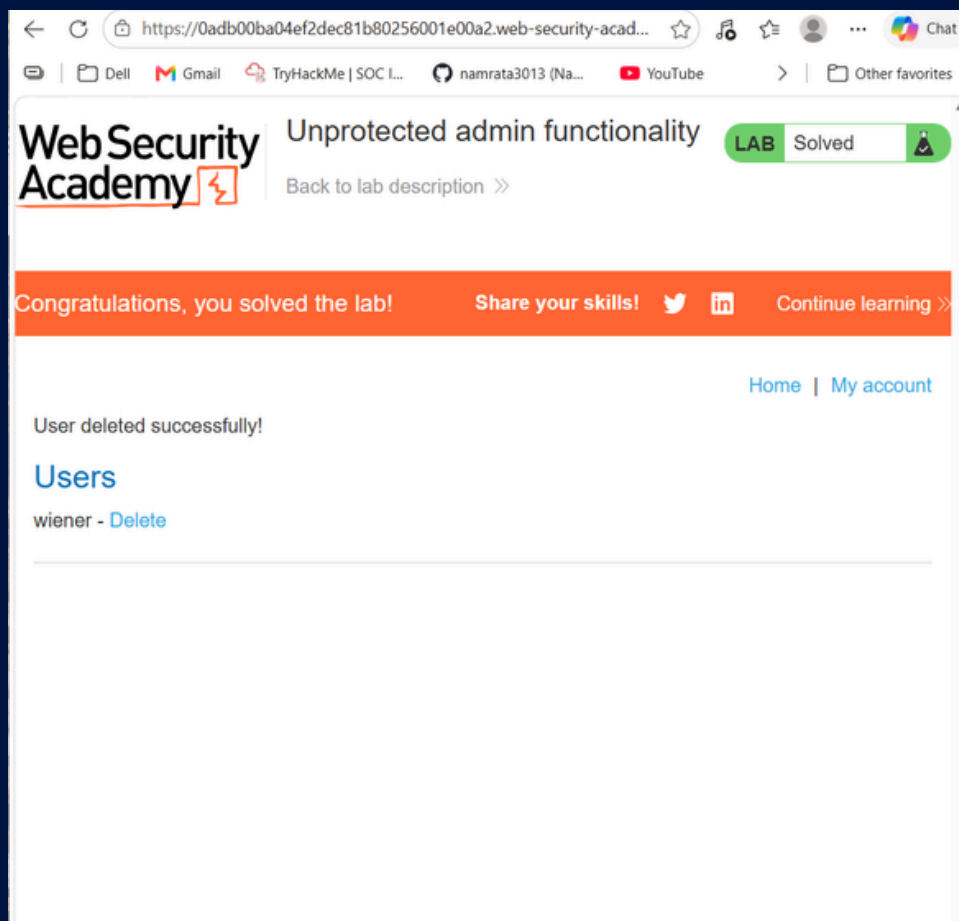
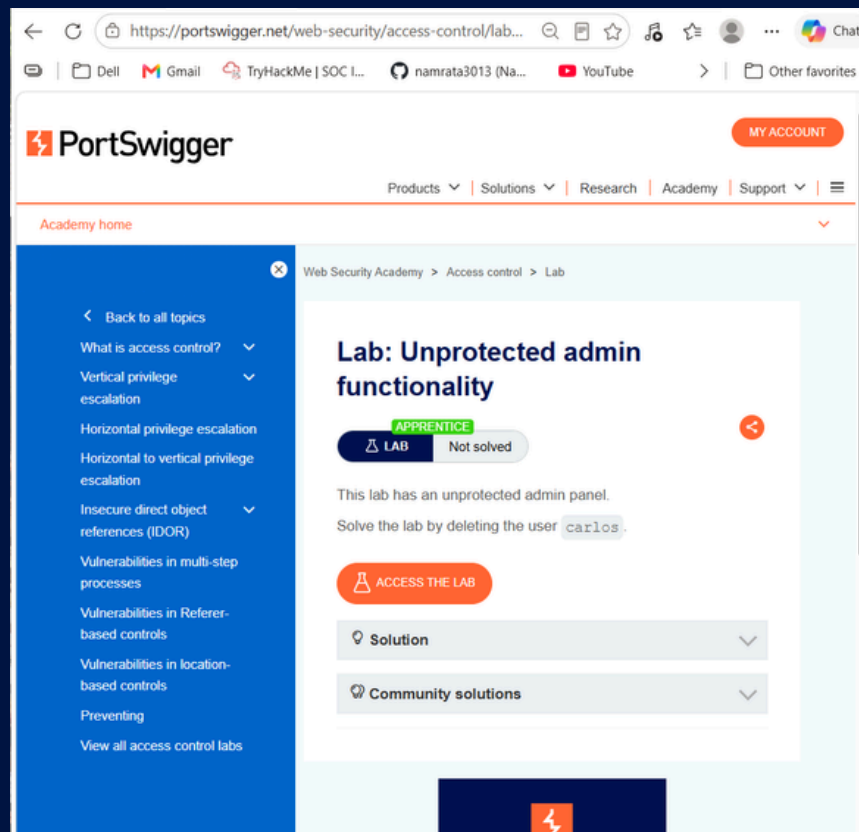
Roulette Drinking Game

Lab 3 — Reflected Cross-Site Scripting (XSS)

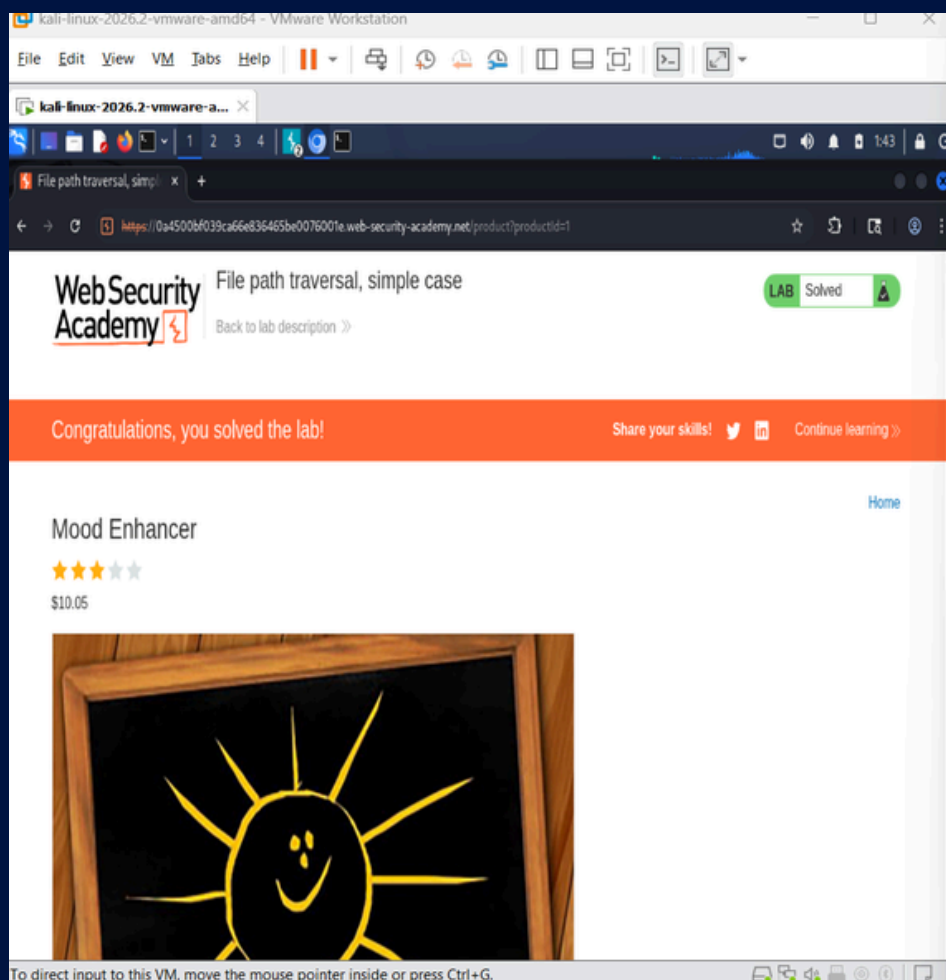
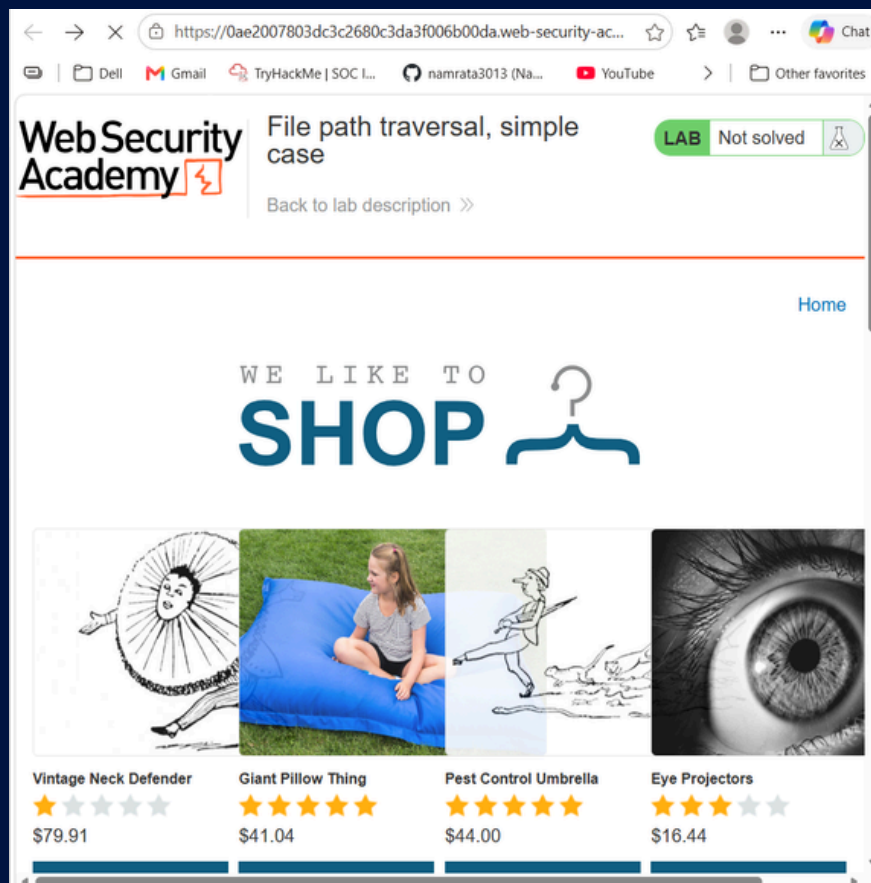
The screenshot shows the PortSwigger Web Security Academy interface. The browser address bar displays `https://portswigger.net/web-security/cross-site-scripting...`. The page title is "Lab: Reflected XSS into HTML context with nothing encoded". The lab is categorized as "APPRENTICE" and "LAB". It is currently marked as "Not solved". The description states: "This lab contains a simple reflected cross-site scripting vulnerability in the search functionality. To solve the lab, perform a cross-site scripting attack that calls the `alert` function." There is an "ACCESS THE LAB" button. Below the description, there are sections for "Solution" and "Community solutions". A left sidebar contains a navigation menu with various topics related to XSS.

The screenshot shows the Web Security Academy search results page. The browser address bar displays `https://0a70003b044e0546823c1fab00d200b4.web-security-acad...`. The page title is "Reflected XSS into HTML context with nothing encoded". The lab is now marked as "LAB Solved". A green banner at the top says "Congratulations, you solved the lab! Share your skills! Continue learning >>". Below the banner, there is a search bar with the text "0 search results for "" and a "Search" button. The search input field contains the payload `<script>alert(1)</script>`. A link to "Back to Blog" is visible at the bottom right.

Lab 4 – Unprotected Admin Functionality



Lab 5 — File Path Traversal



Summary

The SQL Injection exercises demonstrated how improperly handled user input can alter database queries and result in unauthorized authentication or access to data. PortSwigger notes that SQL Injection can allow attackers to interfere with application database queries and potentially retrieve or modify data.

The XSS exercise demonstrated how user-controlled input can be reflected into an HTML response and interpreted as executable browser content.

The unprotected administrative functionality exercise demonstrated the security risk of exposing privileged functionality without appropriate access controls.

The File Path Traversal exercise demonstrated how manipulating a file-related parameter can allow unauthorized access to files on the server. PortSwigger describes path traversal as a vulnerability that can enable attackers to read arbitrary files from the application server.

Each lab was successfully completed, and screenshots were captured as evidence of successful exploitation and lab completion.

Five Apprentice-level web security labs were successfully completed as part of this project.

Lab	Vulnerability	Result
Lab 1	SQL Injection – Login Bypass	Successfully exploited
Lab 2	SQL Injection – Retrieval of Hidden Data	Successfully exploited
Lab 3	Reflected Cross-Site Scripting (XSS)	Successfully exploited
Lab 4	Unprotected Admin Functionality	Successfully exploited
Lab 5	File Path Traversal	Successfully exploited

Conclusion

The PortSwigger Web Security Academy project provided practical experience in identifying, exploiting, and understanding common web application vulnerabilities. Five Apprentice-level labs were successfully completed, covering SQL Injection, Reflected XSS, Unprotected Administrative Functionality, and File Path Traversal.

The exercises demonstrated that improper input handling, insufficient access controls, and insecure interaction with databases or filesystem resources can expose web applications to significant security risks.

The project also provided hands-on experience with Burp Suite for analyzing and modifying HTTP requests. The successful completion of the labs helped strengthen practical understanding of web application security testing and the importance of implementing appropriate defensive controls.

Overall, the project fulfilled its objective of providing practical exposure to web vulnerability assessment in a controlled and authorized environment. Screenshots and lab completion evidence have been retained as supporting proof for the technical report.